AgriSaathi: A Comprehensive Survey on Large Language Models and Multimodal AI for Agricultural Advisory Systems

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Abstract

The agricultural sector faces unprecedented chal- lenges in meeting global food security demands while addressing climate change, resource constraints, and knowledge dissemina- tion gaps. AgriSaathi represents a paradigm shift in agricultural advisory systems by leveraging Large Language Models (LLMs), multimodal AI, and Retrieval-Augmented Generation (RAG) to provide personalized, context-aware guidance to farmers. This survey paper comprehensively examines the evolution of agricultural advisory systems, from traditional extension services to modern AI-driven platforms. We analyze the role of LLMs in transforming agricultural knowledge dissemination, explore multimodal integration of vision, speech, and language processing for crop disease detection and farmer interaction, and investigate RAG frameworks for enhancing domain-specific agricultural knowledge retrieval. The AgriSaathi framework integrates a multilingual LLM-RAG core and vision-based crop diagnosis achieving up to 93–97 accuracy in agricultural query response and disease detection, based on results from benchmark models such as AgroLLM and Krishi Saathi.The paper critically eval- uates existing systems, identifies technical and socio-economic challenges including low-resource language support and digi- tal literacy barriers, and proposes future research directions. AgriSaathi synthesizes these technologies into a unified platform designed specifically for Indian farmers, addressing language diversity, regional crop variations, and real-time environmental data integration. Our findings indicate that LLM-based multi- modal systems can significantly improve agricultural productivity and farmer decision-making when properly adapted to local contexts.

Keywords: GPT, LLM, RAG, Deep Learning, Machine Learning, Smart Agriculture

INTRODUCTION

Agriculture remains the backbone of economies world- wide, employing approximately 26% of the global workforce and contributing significantly to food security and economic development. However, smallholder farmers, particularly in developing nations like India, face multifaceted challenges including limited access to expert agricultural advice, unpre- dictable weather patterns, pest infestations, soil degradation, and market price volatility. Traditional agricultural extension systems, while historically valuable, suffer from scalability issues, delayed information dissemination, and inability to provide personalized recommendations based on real-time farm conditions. The digital revolution and recent advances in artificial intelligence present unprecedented opportunities to democratize agricultural

knowledge and empower farmers with timely, accurate, and context-specific guidance.
Large Language Models (LLMs) have demonstrated re- markable capabilities in natural language understanding, rea- soning, and knowledge synthesis across diverse domains. When applied to agriculture, LLMs can process vast amounts of agricultural literature, government advisories, research pa- pers, and traditional farming knowledge to generate human- like responses to farmer queries. However, agriculture presents unique challenges for LLM deployment: the need for mul- timodal input processing (text, images, voice), support for low-resource regional languages, integration of real-time en- vironmental data, and grounding responses in verified agricul- tural science rather than generic web knowledge. AgriSaathi addresses these challenges by combining LLMs with com- puter vision for crop disease identification, speech recognition for voice-based queries in local languages, and Retrieval-Augmented Generation for accessing curated agricultural knowledge bases.
The emergence of Generative Pre-trained Transformers (GPT) and similar foundation models has revolutionized the landscape of artificial intelligence applications across indus- tries. In agriculture, these models offer transformative potential by enabling natural language interactions between farmers and intelligent advisory systems, democratizing access to expert knowledge that was previously available only through costly consultations or limited extension services. The integration of deep learning techniques, particularly convolutional neural networks for image analysis and recurrent architectures for sequential decision-making, has enabled sophisticated crop disease detection, yield prediction, and resource optimization systems that can operate at scale.
Smart agriculture, characterized by the integration of In- ternet of Things (IoT) sensors, satellite imagery, drone tech- nology, and artificial intelligence, represents a paradigm shift from traditional farming practices. Machine learning algo- rithms analyze multidimensional data streams including soil moisture levels, weather patterns, crop health indicators, and historical yield data to provide actionable insights. However, the true potential of smart agriculture can only be realized when these technological capabilities are made accessible to farmers through intuitive, user-friendly interfaces that over- come barriers of language, literacy, and digital infrastructure. This is where multimodal AI systems, powered by Large Lan- guage Models and Retrieval-Augmented Generation, become crucial.
The development of AgriSaathi is motivated by the specific needs of Indian agriculture, where extreme diversity in crops, climatic zones, languages, and farming practices demands highly adaptable and context-aware advisory systems. India’s agricultural landscape spans from the rice paddies of West Bengal to the cotton fields of Gujarat, from the coffee planta- tions of Karnataka to the wheat belts of Punjab, each requiring specialized knowledge and localized recommendations. Tradi- tional one-size-fits-all approaches fail to address this hetero- geneity, while human extension services cannot scale to serve over 140 million farming households. AgriSaathi leverages the generalization capabilities of LLMs, combined with region- specific knowledge bases accessed through RAG, to provide personalized guidance that accounts for local conditions, crop varieties, pest pressures, and market dynamics.
This survey paper provides a comprehensive analysis of AI- driven agricultural advisory systems with specific focus on LLM-based approaches. We examine the historical evolution of agricultural extension services, review state-of-the-art re- search in agricultural chatbots and decision support systems, analyze the technical architecture of multimodal AI platforms, and investigate RAG frameworks for knowledge-intensive agricultural applications. Section II provides background on agricultural advisory systems and reviews related work in AI for agriculture. Section III presents a comprehensive literature survey of recent advances in agricultural AI systems. Section IV examines the transformative role of LLMs in

agricultural applications. Section V investigates multimodal AI integration for vision, speech, and language processing. Section VI ana- lyzes RAG methodologies for agricultural knowledge systems. Section VII discusses challenges and limitations, Section VIII identifies key research gaps, and Section IX concludes with insights on AgriSaathi's potential impact on Indian agriculture, followed by acknowledgements.

Background and Related Work

Agricultural advisory systems have evolved significantly from traditional face-to-face extension services to digital plat- forms leveraging mobile technology and artificial intelligence.

Early digital agricultural systems primarily focused on de- livering static information through SMS services and basic mobile applications, providing weather updates, market prices, and generic farming tips. However, these systems lacked personalization and could not address the diverse, context- specific challenges faced by individual farmers across different agro-climatic zones. The introduction of machine learning techniques in the 2010s enabled more sophisticated decision support systems that could predict crop yields, detect diseases from images, and recommend optimal fertilizer applications based on soil conditions.

Recent advances in deep learning and natural language processing have catalyzed a new generation of intelligent agri- cultural systems. Convolutional Neural Networks (CNNs) have achieved remarkable accuracy in crop disease identification from leaf images, with some studies reporting over 95% clas- sification accuracy for common diseases. Simultaneously, the development of pre-trained language models like BERT and GPT has enabled more natural conversational interfaces for agricultural information systems. Researchers have explored domain-specific adaptations of these models for agriculture, including fine-tuning on agricultural corpora and developing specialized embeddings for crop names, disease terminology, and farming practices. These developments have paved the way for comprehensive platforms like AgriSaathi that integrate multiple AI technologies into unified advisory systems.

The literature reveals several notable agricultural chatbot implementations worldwide. Kisan Suvidha in India provides weather-based advisories and market information through a mobile application, while Plantix combines image recognition with a knowledge base to diagnose plant diseases and sug- gest treatments. Research initiatives such as the deployment of voice-based agricultural advisory systems in Africa have demonstrated the importance of supporting oral communica- tion for farmers with limited literacy. However, most existing systems operate in isolation, focusing on specific aspects like disease detection or market information rather than provid- ing holistic farm management advice. Furthermore, studies have identified critical gaps in supporting regional languages, integrating real-time sensor data from farms, and providing explanations for AI-generated recommendations that farmers can understand and trust.

Literature Survey

Recent advances in artificial intelligence have spawned numerous research efforts aimed at revolutionizing agricultural practices through smart technologies. Khaliq et al. [1] pre- sented an AI-driven smart agriculture approach that integrates soil analysis, irrigation management, and crop-fertilizer rec- ommendations, proposing a comprehensive system that lever- ages machine learning algorithms to analyze soil properties and environmental conditions. Their work demonstrates how integrated AI approaches can optimize resource utilization in farming operations. The proposed methodology combines multiple data sources including soil sensors and weather information, though limitations include the need for extensive sensor infrastructure and potential challenges in scaling to smallholder farms. The research gap identified relates to the integration of farmer experiential knowledge alongside sensor data for more holistic decision-making. Yang et al. [2] intro- duced AgriGPT, a large language model ecosystem specifically designed for agricultural applications, developing domain- adapted GPT models fine-tuned on agricultural corpora to handle farmer queries and provide expert guidance. Their approach demonstrates the potential of foundation models in agriculture, with proposed methods including agricultural corpus creation and domain-specific fine-tuning strategies. However, limitations include language constraints primarily focusing on high-resource languages and potential hallucina- tion issues in specialized agricultural contexts. The research gap involves developing more robust grounding mechanisms and expanding multilingual capabilities for diverse farming communities. Jiang et al. [3] developed Farm-LightSeek, an edge-centric multimodal agricultural IoT framework with lightweight LLMs, addressing the computational constraints of edge deployment while maintaining analytical capabilities. Their proposed system enables on-device processing of multi- modal agricultural data including images, sensor readings, and text, using model compression and quantization techniques to deploy LLMs on resource-constrained hardware. Limitations include reduced model capacity compared to full-scale LLMs and potential accuracy trade-offs. Research gaps exist in bal- ancing model size with performance and developing efficient update mechanisms for edge-deployed models.

Siino et al. [4] explored LLM applications in law through a comprehensive literature review, providing insights into how similar NLP approaches could be adapted for agricultural reg- ulatory compliance and policy navigation. Their work in legal NLP reveals methodological parallels applicable to agricul- ture, particularly in handling domain-specific terminology and regulatory frameworks. The proposed approaches include spe- cialized fine-tuning and retrieval methods, though limitations center on domain transfer challenges. Research gaps include developing agricultural-legal knowledge integration systems that help farmers understand and comply with complex agri- cultural policies. Hindi et al. [5] surveyed retrieval-augmented generation techniques in legal technology, examining how RAG can enhance precision and interpretability in knowledge- intensive domains, findings directly applicable to agricultural knowledge systems. Their comprehensive analysis of RAG architectures, retrieval mechanisms, and generation strategies provides a blueprint for agricultural implementations. Pro- posed methods include hybrid retrieval combining dense and sparse techniques, with limitations in handling multilingual retrieval and dynamic knowledge updates. Research gaps involve developing agricultural-specific RAG systems with temporal awareness and regional knowledge organization. A comprehensive review on enhancing agricultural intelligence with large language models [6] systematically analyzed the state-of-the-art in applying LLMs to various agricultural tasks including crop management, disease diagnosis, and farmer advisory services. The review identifies key challenges in adapting general-purpose LLMs to agricultural domains and proposes future research directions. Limitations noted include data scarcity for specialized crops and the need for better evaluation frameworks specific to agricultural applications. Research gaps encompass developing standardized bench- marks for agricultural LLMs and creating comprehensive multilingual agricultural corpora.

Friha et al. [7] conducted an extensive survey on LLM- based edge intelligence, covering architectures, applications, security, and trustworthiness aspects crucial for agricultural deployments where edge processing is essential. Their work addresses the unique constraints and requirements of de- ploying

LLMs in resource-limited edge environments typi- cal of farm settings. Proposed approaches include federated learning and model partitioning strategies, though limita- tions exist in maintaining model performance under strict resource constraints. Research gaps include developing trust frameworks for edge-deployed agricultural AI and ensuring data privacy in distributed farm networks. Saha et al. [8] advanced retrieval-augmented generation with their QuIM- RAG approach using inverted question matching to enhance question-answering performance, demonstrating improved re- trieval precision through query reformulation techniques appli- cable to agricultural knowledge retrieval. Their methodology shows how better query understanding can improve RAG system effectiveness, with proposed methods including query expansion and multi-perspective retrieval. Limitations involve computational overhead of query processing and dependency on query quality. Research gaps relate to adapting these techniques for voice-based agricultural queries and handling dialectical variations in farmer questions. Yenduri et al. [9] provided a comprehensive review of GPT technology, exam- ining enabling technologies, potential applications, emerging challenges, and future directions across domains including agriculture. Their analysis covers the full spectrum of GPT capabilities and limitations, offering insights into both op- portunities and risks of deploying such models in critical applications. Proposed applications span multiple agricultural tasks, though limitations include energy consumption, com- putational requirements, and ethical considerations. Research gaps involve developing sustainable deployment models and addressing bias in agricultural recommendations.

Zhu et al. [10] surveyed the use of large vision and language models in agriculture, examining how multimodal foundation models can process both visual and textual agricultural in- formation for comprehensive analysis. Their review highlights advances in vision-language pre-training and applications to crop monitoring, disease detection, and agricultural robotics. Proposed methods integrate visual encoders with language models, though limitations include data requirements for fine- tuning and challenges in handling low-quality field images. Research gaps exist in developing robust multimodal models that perform well with real-world agricultural imagery cap- tured under varying conditions. Al-Shahari et al. [11] explored IoT-assisted plant disease detection and crop management using deep learning for sustainable agriculture, proposing integrated systems that combine sensor networks with computer vision models. Their approach demonstrates how continuous monitoring through IoT devices coupled with deep learning analysis can enable early disease detection and preventive interventions. Proposed methods include real-time disease classification pipelines, with limitations in sensor deployment costs and connectivity requirements. Research gaps involve developing cost-effective IoT solutions for smallholder farmers and ensuring system reliability under field conditions. A study on large language models and foundation models in smart agriculture [12] examined how these models can transform agricultural decision-making, knowledge management, and automation, analyzing various architectures and their suit- ability for different agricultural tasks. The research identifies opportunities in areas such as automated advisory services, literature synthesis, and farmer education. Limitations in- clude the need for agricultural domain expertise to guide model development and challenges in evaluation. Research gaps encompass creating comprehensive agricultural reasoning benchmarks and developing explainable AI methods tailored to farming contexts.

Balafas et al. [13] conducted a thorough review of ma- chine learning and deep learning approaches for plant disease classification and detection, comparing various neural network architectures and analyzing their performance across different crops and diseases. Their work provides detailed insights into model selection, dataset requirements, and deployment con- siderations for practical disease detection systems.

Proposed methods span traditional machine learning to state-of-the-art deep learning, with limitations including dataset imbalance and generalization challenges across crop varieties. Research gaps involve developing few-shot learning approaches for rare diseases and creating standardized evaluation protocols. Elbasi et al. [14] presented a systematic literature review of artificial intelligence technology in the agricultural sector, comprehen- sively cataloging AI applications across the agricultural value chain from land preparation to post-harvest processing. Their systematic approach identifies trends, research gaps, and future opportunities in agricultural AI. Proposed technologies cover the full spectrum of AI techniques, though limitations noted include adoption barriers and infrastructure requirements. Research gaps relate to developing practical implementation frameworks and addressing socio-economic factors affecting technology adoption. Thakur et al. [15] developed PlantXViT, an explainable vision transformer enabled convolutional neural network for plant disease identification, addressing the critical need for interpretability in AI-driven agricultural diagnostics. Their hybrid architecture combines the strengths of CNNs and vision transformers while providing visual explanations for predictions through attention mechanisms. Proposed methods generate saliency maps highlighting disease-affected regions, though limitations include computational complexity and the need for expert validation of explanations. Research gaps exist in developing real-time explainable models and creating standardized metrics for explanation quality in agricultural contexts.

Li et al. [16] provided an extensive review of plant disease detection and classification through deep learning, analyzing the evolution from early CNNs to sophisticated attention- based architectures and examining challenges in practical deployment. Their comprehensive analysis covers dataset cre- ation, model architectures, training strategies, and performance evaluation across diverse agricultural scenarios. Proposed ap- proaches include transfer learning and data augmentation techniques, with limitations in handling environmental vari- ability and disease progression stages. Research gaps involve developing temporal disease progression models and creating comprehensive multi-disease, multi-crop datasets. Roy et al.

[17] presented a fast accurate fine-grain object detection model based on YOLOv4 deep neural network, demonstrating real- time detection capabilities crucial for practical agricultural applications requiring immediate feedback. Their optimized architecture achieves balance between detection accuracy and inference speed, making it suitable for deployment on mobile devices and edge hardware in farm settings. Proposed methods include model compression and efficient feature extraction, though limitations exist in detecting small objects and handling occlusions. Research gaps relate to multi-scale detection in complex agricultural scenes and robust performance under varying lighting conditions. Huang et al. [19] introduced VLLFL, a vision-language model based lightweight feder- ated learning framework for smart agriculture, addressing privacy concerns and communication constraints in distributed agricultural systems. Their framework enables collaborative learning across multiple farms while preserving data pri- vacy through federated approaches combined with lightweight vision-language models. Proposed methods include efficient aggregation strategies and model compression techniques, with limitations in heterogeneous data handling and communication overhead. Research gaps involve developing adaptive federated learning methods that account for varying farm conditions and ensuring model fairness across diverse agricultural contexts.

Role of Large Language Models in Agriculture

Large Language Models have revolutionized natural lan- guage processing through their ability to understand context, generate coherent text, and perform reasoning tasks across diverse domains. Models such as GPT-4, Claude, and LLaMA demonstrate impressive capabilities in question answering, text summarization, and multi-turn dialogue that can be directly ap- plied to agricultural advisory systems. Unlike traditional rule- based chatbots that follow predefined decision trees, LLMs can handle open-ended farmer queries, provide nuanced ex- planations, and adapt their responses based on conversational context. For AgriSaathi, LLMs serve as the core conversa- tional engine, processing farmer questions in natural language and generating informative, actionable responses grounded in agricultural knowledge.

The application of LLMs to agriculture requires care- ful consideration of domain-specific challenges. General- purpose LLMs trained primarily on web text may generate plausible-sounding but agriculturally incorrect advice, a phe- nomenon known as hallucination. To address this, researchers have explored various adaptation strategies including fine- tuning LLMs on agricultural corpora, implementing retrieval- augmented generation to ground responses in verified sources, and developing specialized prompting techniques that encour- age factual, conservative responses. Agricultural LLMs must also handle diverse regional contexts, as farming practices vary significantly based on soil type, climate, crop varieties, and local traditions. Studies have shown that incorporating local agricultural knowledge bases and region-specific crop calendars significantly improves the relevance and accuracy of LLM-generated advice.

Language diversity represents a critical consideration for agricultural LLM deployment in multilingual countries like In- dia, where farmers speak hundreds of regional languages with varying levels of digital representation. While LLMs perform exceptionally well in high-resource languages like English, Hindi, and Mandarin, their performance degrades significantly in low-resource languages common in rural agricultural com- munities. Researchers have investigated cross-lingual transfer learning, where models trained on high-resource languages are adapted to low-resource languages through techniques like multilingual pre-training and translation-based augmentation. AgriSaathi leverages multilingual LLMs with specialized fine- tuning on agricultural terminology in regional Indian lan- guages, enabling farmers to interact in their native languages while receiving accurate, context-appropriate guidance. Ad- ditionally, the integration of automatic speech recognition allows voice-based interaction, crucial for farmers with limited literacy who may struggle with text-based interfaces.

MULTIMODAL AI: VISION, SPEECH, AND LANGUAGE INTEGRATION

Multimodal AI systems that process and integrate infor- mation from multiple input modalities—text, images, audio, and sensor data—offer significant advantages for agricultural applications where farmers' problems often cannot be ade- quately expressed through text alone. Crop disease diagnosis exemplifies the necessity of multimodal processing: a farmer may struggle to describe symptoms accurately in words, but a photograph of affected leaves provides immediate visual evidence that computer vision models can analyze. Similarly, voice-based interaction proves essential for farmers who are illiterate or uncomfortable with text input, while environmental sensor data provides objective measurements that comple- ment farmer-reported observations. AgriSaathi integrates these modalities into a unified system where information flows seamlessly between vision, speech, and language processing components.

Computer vision plays a pivotal role in modern agricultural advisory systems, particularly for crop disease detection and pest identification. Deep convolutional neural networks trained on extensive image datasets can identify dozens of plant diseases with accuracy rivaling or exceeding human experts. State-of-the-art models like ResNet, EfficientNet, and Vision Transformers have been successfully applied to agricultural image classification tasks, processing images captured by smartphone cameras to detect diseases, nutrient deficiencies, and pest infestations. For AgriSaathi, the vision module pro- cesses farmer-submitted images of crops, leaves, or insects, extracting visual features that are then contextualized by the LLM to generate comprehensive diagnostic reports with treat- ment recommendations. Recent advances in few-shot learning and transfer learning enable these systems to recognize rare diseases or new crop varieties with minimal training examples, addressing the long-tail distribution problem in agricultural datasets.

Speech recognition and synthesis technologies enable nat- ural voice-based interaction with agricultural advisory sys- tems, crucial for accessibility and user experience. Automatic Speech Recognition (ASR) systems must handle challenges specific to agricultural contexts: background noise from farm environments, accented speech from diverse regional pop- ulations, and agricultural terminology that may not appear in standard speech recognition vocabularies. Modern ASR systems based on transformer architectures and end-to-end learning approaches have achieved significant improvements in multilingual recognition, though performance still varies considerably across languages. Text-to-Speech (TTS) synthe- sis allows the system to deliver responses vocally, enabling hands-free operation while farmers work in fields. AgriSaathi implements multilingual speech interfaces supporting major Indian languages, with acoustic models adapted to rural speech patterns and agricultural vocabulary. The integration of speech, vision, and language creates a truly multimodal experience where farmers can photograph a diseased plant, describe symptoms verbally, and receive spoken guidance—all within a single conversational flow.

Retrieval-Augmented Generation for Agricultural Knowledge Systems

Retrieval-Augmented Generation represents a paradigm shift in addressing the knowledge grounding challenge for LLMs in specialized domains like agriculture. Rather than relying solely on knowledge encoded in model parameters dur- ing pre-training, RAG systems augment the generation process by retrieving relevant information from external knowledge bases before formulating responses. For agricultural appli- cations, this approach offers critical advantages: responses can be grounded in verified agricultural research, government guidelines, and local best practices; the system can access up-to-date information without requiring model retraining; and sources can be cited to build farmer trust. AgriSaathi employs RAG architecture where farmer queries trigger retrieval of relevant documents from curated agricultural knowledge bases, which then inform the LLM's response generation.

The effectiveness of RAG systems depends critically on the quality, organization, and retrieval mechanisms of the un- derlying knowledge base. For agriculture, relevant knowledge sources include scientific research papers, agricultural exten- sion bulletins, crop cultivation handbooks, pest management guides, government schemes and subsidies, and experiential knowledge from successful farmers. These documents must be processed, indexed, and organized to enable efficient retrieval based on semantic similarity to farmer queries. Dense retrieval methods utilizing neural embeddings have largely superseded traditional keyword-based search, with models like Dense Pas- sage Retrieval (DPR) and Contriever demonstrating superior performance in finding relevant documents based on semantic meaning rather than lexical overlap. AgriSaathi implements a hierarchical knowledge base organized by crop type, agro- climatic zone, and problem category, with specialized retrieval models fine-tuned on agricultural query-document pairs to improve retrieval precision.

Recent advances in RAG methodology address key chal- lenges in agricultural knowledge systems.

Hybrid retrieval approaches combine dense semantic retrieval with sparse keyword matching to handle both conceptual queries and specific entity mentions like crop varieties or chemical names. Multi-hop reasoning capabilities enable the system to retrieve and synthesize information from multiple documents when answering complex queries that require connecting informa- tion from different sources. Temporal awareness ensures that seasonally relevant information receives higher priority—for example, prioritizing monsoon preparation advice during pre- monsoon months. AgriSaathi also incorporates real-time data integration, where RAG retrieval is augmented with current weather forecasts, market prices, and pest outbreak alerts from government databases. This fusion of static knowledge re- trieval with dynamic data streams creates a comprehensive in- formation foundation that enables the LLM to generate timely, accurate, and actionable agricultural guidance grounded in both scientific knowledge and current conditions.

Challenges and Limitations

Despite significant technological advances, deploying LLM- based agricultural advisory systems like AgriSaathi faces sub- stantial technical, socio-economic, and practical challenges. Data availability and quality remain fundamental obstacles: training robust agricultural AI systems requires extensive datasets of crop images, disease annotations, voice samples in regional languages, and agricultural text corpora. Unlike domains with abundant digital data, agriculture—particularly in developing countries—suffers from limited digitized knowl- edge, fragmented data across institutions, and scarcity of labeled datasets for supervised learning. Regional variations in crops, farming practices, and pest populations necessitate lo- calized datasets that often do not exist or are prohibitively ex- pensive to create. Furthermore, agricultural knowledge evolves with climate change, introduction of new crop varieties, and emergence of novel pests, requiring continuous data collection and model updating.

Language diversity and digital literacy present critical ac- cessibility challenges. India alone has 22 officially recog- nized languages and hundreds of dialects, many of which are low-resource languages with limited digital corpora, speech recognition models, or LLM support. Developing high-quality multimodal AI systems for all relevant agricultural languages requires substantial investment in data collection, model train- ing, and interface localization. Beyond language, digital lit- eracy varies significantly among farming communities, with older farmers and those in remote areas often unfamiliar with smartphone applications and conversational interfaces. System design must accommodate varying levels of technical proficiency through intuitive interfaces, voice-based interaction as an alternative to text, and offline functionality for areas with unreliable internet connectivity. Additionally, trust and adoption barriers exist where farmers may be skeptical of AI- generated advice, preferring traditional knowledge or human experts.

Technical limitations of current AI systems pose risks when deployed in high-stakes agricultural contexts. LLMs can gen- erate incorrect or outdated information (hallucination), poten- tially leading farmers to make harmful decisions regarding pes- ticide application, fertilizer use, or crop management. Ensuring safety requires robust validation mechanisms, conservative response generation, and clear communication of uncertainty when the system lacks confidence in its recommendations. Computer vision models may misclassify diseases, particu- larly when image quality is poor or diseases present atypical symptoms. Infrastructure constraints including limited inter- net bandwidth, unreliable electricity, and lack of smartphone ownership among target users further complicate deployment. Privacy and data security concerns arise when systems collect sensitive farm data, requiring careful implementation of data protection measures. Addressing these challenges necessitates interdisciplinary collaboration between AI researchers, agri- cultural scientists, extension officers, and farming communities to develop systems that are technically robust, culturally appropriate, and genuinely beneficial to farmers.

Research Gaps

The future evolution of LLM-based agricultural advisory systems like AgriSaathi lies in several promising research directions that can enhance functionality, accessibility, and impact. Personalization and context-awareness represent key opportunities: future systems should maintain comprehensive farmer profiles including land holdings, cropping patterns, his- torical yields, equipment availability, and financial constraints to provide truly customized recommendations. Integration with IoT sensor networks deployed in fields can enable real-time monitoring of soil moisture, nutrient levels, and microclimate conditions, allowing the LLM to generate dynamic advice re- sponsive to actual farm conditions rather than generic regional data. Advanced reasoning capabilities through techniques like chain-of-thought prompting and tool-augmented LLMs can enable more sophisticated problem-solving, such as multi-season crop rotation planning that optimizes soil health, pest management, and economic returns.

Expanding multimodal capabilities beyond current vision and speech processing offers significant potential. Satellite and drone imagery analysis can provide farm-scale assessments of crop health, growth patterns, and stress indicators, com- plementing ground-level smartphone photos. Spectral imaging techniques using hyperspectral or multispectral cameras can detect disease or nutrient deficiencies before visible symp- toms appear, enabling preventive interventions. Integration with market intelligence systems and supply chain platforms can help farmers make informed decisions about what to plant, when to harvest, and where to sell for optimal returns. Social connectivity features enabling farmer-to-farmer knowl- edge sharing, peer learning networks, and community-based problem-solving can leverage collective intelligence alongside AI guidance. Additionally, augmented reality interfaces could provide visual overlays showing optimal planting layouts, irrigation patterns, or step-by-step guidance for agricultural procedures.

Research priorities should address fundamental limitations in current approaches. Developing more robust, agriculture- specific LLMs through pre-training on extensive agricultural corpora and fine-tuning on diverse farmer interaction data can improve domain expertise and reduce hallucination. Ad- vancing low-resource language technologies through transfer learning, synthetic data generation, and community-driven data collection can expand accessibility. Explainable AI techniques that provide transparent reasoning for recommendations can build farmer trust and enable learning from the system's advice. Federated learning approaches can enable collaborative model improvement across multiple regions while preserving data privacy. Long-term impact studies evaluating effects on farmer income, crop yields, environmental sustainability, and knowledge retention are essential to validate effectiveness and guide iterative improvements. Finally, developing sustainable business models and deployment strategies that ensure af- fordability and accessibility for smallholder farmers remains crucial for achieving scale and impact.

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Conclusion

This survey has comprehensively examined the landscape of LLM-based agricultural advisory systems with focus on AgriSaathi, a multimodal AI platform designed to empower Indian farmers with personalized, timely, and accurate agri- cultural guidance. We have traced the evolution from tra- ditional extension services through digital information sys- tems to contemporary AI-driven platforms that leverage large language models, computer vision, and speech processing. The analysis demonstrates that while significant technological advances have created unprecedented opportunities for intel- ligent agricultural advisory services, successful deployment requires careful attention to domain-specific challenges in- cluding knowledge grounding, language diversity, multimodal integration, and socio-economic accessibility constraints.

References

1. A. Khaliq et al., “AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis, Irrigation, and Crop-Fertilizer Recommendations,” *IEEE Access*, vol. 13, 2025.
2. B. Yang et al., “AgriGPT: A Large Language Model Ecosystem for Agriculture,” vol. 17, 2025.
3. D. Jiang et al., “Farm-LightSeek: An Edge-centric Multimodal Agricul- tural IoT Data Analytics Framework with Lightweight LLMs,” vol. 13, 2025.
4. M. Siino et al., “Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches,” *IEEE Access*, vol. 13, 2025.
5. M. Hindi et al., “Enhancing the Precision and Interpretability of Retrieval-Augmented Generation (RAG) in Legal Technology: A Sur- vey,” *IEEE Access*, vol. 13, 2025.
6. “A review on enhancing agricultural intelligence with large language models,” ScienceDirect article, vol. 17, 2025.
7. O. Friha et al., “LLM-Based Edge Intelligence: A Comprehensive Survey on Architectures, Applications, Security and Trustworthiness,” *IEEE Open Journal of the Communications Society*, vol. 5, 2024.
8. B. Saha et al., “QuIM-RAG: Advancing Retrieval-Augmented Genera- tion With Inverted Question Matching for Enhanced QA Performance,” *IEEE Access*, vol. 12, 2024.
9. G. Yenduri et al., “GPT (Generative Pre-Trained Transformer)—A Com- prehensive Review on Enabling Technologies, Potential Applications, Emerging Challenges, and Future Directions,” *IEEE Access*, vol. 12, 2024.
10. H. Zhu et al., “Harnessing Large Vision and Language Models in Agriculture: A Review,” vol. 11, 2024.
11. E. Al-Shahari et al., “Internet of Things Assisted Plant Disease Detection and Crop Management Using Deep Learning for Sustainable Agricul- ture,” *IEEE Access*, vol. 15, 2024.
12. “Large Language Models and Foundation Models in Smart Agriculture,” vol. 18, 2024.
13. V. Balafas et al., “Machine Learning and Deep Learning for Plant Disease Classification and

Detection," *IEEE Access*, vol. 11, 2023.
14. E. Elbasi et al., "Artificial Intelligence Technology in the Agricultural Sector: A Systematic Literature Review," *IEEE Access*, vol. 11, 2023.
15. Poornima Singh Thakur et al., "Explainable Vision Transformer En- abled Convolutional Neural Network for Plant Disease Identification: PlantXViT," vol. 14, 2022.
16. L. Li et al., "Plant Disease Detection and Classification by Deep Learning—A Review," *IEEE Access*, vol. 9, 2021.
17. A. M. Roy et al., "A Fast Accurate Fine-Grain Object Detection Model Based on YOLOv4 Deep Neural Network," vol. 10, 2021.
18. D. Jiang et al., "Farm-LightSeek: An Edge-centric Multimodal Agricul- tural IoT Data Analytics Framework with Lightweight LLMs," vol. 41, 2025.
19. Y. Huang et al., "VLLFL: A Vision-Language Model Based Lightweight Federated Learning Framework for Smart Agriculture," ArXiv preprint, 2025.
20. P. S. Thakur et al., "Explainable Vision Transformer Enabled Convolu- tional Neural Network for Plant Disease Identification: PlantXViT," vol. 11, 2022.